# Xiaoxingkai Transfer Result Report

Run directory: S:\STU-Papers\My_Papers\EXP1-Pre-Training-Standalone\outputs\runs\xiaoxingkai_transfer_daily12\xiaoxingkai_transfer_daily12_20260415_115007
Experiment: xiaoxingkai_transfer_daily12
Source file: S:\STU-Papers\My_Papers\Pre-Training-Data\LakeIce_ERA5_less60_XiaoxingkaiDaily12.xlsx
Target lake: [Li] Lake Xiaoxingkai

## Front Summary

| Split | Loss | RMSE | MAE | R2 | Bias | Negative Preds |
|---|---|---|---|---|---|---|
| val | 0.1109 | 0.3270 | 0.2778 | -0.7052 | 0.2147 | 0 |
| test | 0.1405 | 0.3748 | 0.3105 | -81.3576 | -0.1807 | 0 |

## Per-Lake Numeric Evaluation Summary

| Lake | Count | RMSE | MAE | R2 | Bias |
|---|---|---|---|---|---|
| [Li] Lake Xiaoxingkai | 36 | 0.3748 | 0.3105 | -81.3576 | -0.1807 |

## Key Diagnostic Notes

Best epoch: 42

Observed evaluation mean/std: 0.7567 / 0.0413

Predicted evaluation mean/std: 0.5760 / 0.3283

Validation windows: 41 | Test windows: 36

Interpretation: if predicted std is much smaller than observed std, the model is behaving too conservatively and is under-expressing peak-to-trough variation.

# 0. Training Inputs And Hyperparameters

The model consumes one irregular path per valid anchor row. Each path contains relative time plus meteorological and seasonal channels; the target total_ice_m is predicted, not fed back as an input feature.

### *Input Channels*

1. relative_time
2. doy_sin
3. doy_cos
4. Ten_Meter_Elevation_Wind_Speed_meterPerSecond
5. Air_Temperature_celsius
6. Relative_Humidity_percent
7. Shortwave_Radiation_Downwelling_wattPerMeterSquared
8. Longwave_Radiation_Downwelling_wattPerMeterSquared
9. Sea_Level_Barometric_Pressure_pascal
10. Surface_Level_Barometric_Pressure_pascal
11. Precipitation_millimeterPerDay
12. Snowfall_millimeterPerDay

### *Core Training Configuration*

| Field | Value |
|---|---|
| window.window_days | 10 |
| coeffs.interpolation | hermite |
| model.method | midpoint |
| model.hidden_channels | 48 |
| model.hidden_hidden_channels | 96 |
| model.num_hidden_layers | 2 |

| Field | Value |
|---|---|
| train.batch_size | 64 |
| train.learning_rate | 0.0002 |
| train.weight_decay | 0.0005 |
| train.loss | huber |
| train.huber_delta | 0.1 |
| train.max_epochs | 120 |
| train.gradient_clip_norm | 0.5 |
| custom_split.val_fraction | 0.1 |
| custom_split.min_train_rows_per_lake | 24 |
| custom_split.min_val_rows_per_lake | 8 |
| custom_split.max_train_windows_per_lake | 160 |
| custom_split.max_val_windows_per_lake | 20 |
| custom_split.target_lake_test_start | 2026-01-01 |

# 1. Source-Lake Selection, Windowing, And Balance Effects

Each source lake is first split by time into train/val. The target lake [Li] Lake Xiaoxingkai is split so that rows before the cutoff are further divided into train/val, while rows from the cutoff onward are reserved for test. A row only becomes a valid window if there are at least two observations inside the 10-day lookback. After that, train and val windows are capped per lake for balance.

### *Row Assignment Counts From The Source File*

| Lake | Rows In Source | Train Rows | Val Rows | Test Rows |
|---|---|---|---|---|
| [Li] Lake Xiaoxingkai | 68 | 24 | 7 | 37 |
| [Huo] Lake Hnagzhang | 101 | 90 | 11 | 0 |
| [Huo] Lake Ulansu | 163 | 146 | 17 | 0 |
| [IISD Experimental] Lake 239 | 160 | 144 | 16 | 0 |
| [IISD Experimental] Lake 302 | 101 | 90 | 11 | 0 |
| [North Temperate Lakes LTER] Allequash Lake | 118 | 106 | 12 | 0 |
| [North Temperate Lakes LTER] Big Muskellunge Lake | 121 | 108 | 13 | 0 |
| [North Temperate Lakes LTER] Bog 12-15 (Trout Bog) | 124 | 111 | 13 | 0 |
| [North Temperate Lakes LTER] Bog 27-2 (Crystal Bog) | 121 | 108 | 13 | 0 |
| [North Temperate Lakes LTER] Crystal Lake (NTL) | 120 | 108 | 12 | 0 |
| [North Temperate Lakes LTER] Sparkling Lake | 121 | 108 | 13 | 0 |
| [North Temperate Lakes LTER] Trout Lake | 120 | 108 | 12 | 0 |

### *Window Eligibility And Balance Effects*

| Lake | Train Valid | Train Kept | Val Valid | Val Kept | Test Windows | Median Points |
|---|---|---|---|---|---|---|
| [Li] Lake Xiaoxingkai | 23 | 23 | 7 | 7 | 36 | 11.0 |
| [Huo] Lake Hnagzhang | 89 | 89 | 11 | 11 | 0 | 11.0 |

| Lake | Train Valid | Train Kept | Val Valid | Val Kept | Test Windows | Median Points |
|---|---|---|---|---|---|---|
| [Huo] Lake Ulansu | 144 | 144 | 17 | 17 | 0 | 11.0 |
| [IISD Experimental] Lake 239 | 21 | 21 | 2 | 2 | 0 | 2.0 |
| [IISD Experimental] Lake 302 | 6 | 6 | 3 | 3 | 0 | 2.0 |
| [North Temperate Lakes LTER] Allequash Lake | 1 | 1 | 0 | 0 | 0 | 2.0 |
| [North Temperate Lakes LTER] Big Muskellunge Lake | 0 | 0 | 0 | 0 | 0 | 0.0 |
| [North Temperate Lakes LTER] Bog 12-15 (Trout Bog) | 0 | 0 | 1 | 1 | 0 | 2.0 |
| [North Temperate Lakes LTER] Bog 27-2 (Crystal Bog) | 0 | 0 | 0 | 0 | 0 | 0.0 |
| [North Temperate Lakes LTER] Crystal Lake (NTL) | 0 | 0 | 0 | 0 | 0 | 0.0 |
| [North Temperate Lakes LTER] Sparkling Lake | 0 | 0 | 0 | 0 | 0 | 0.0 |
| [North Temperate Lakes LTER] Trout Lake | 0 | 0 | 0 | 0 | 0 | 0.0 |

### *Total Counts*

Total included source rows: 1438

Total train rows / val rows / test rows: 1251 / 150 / 37

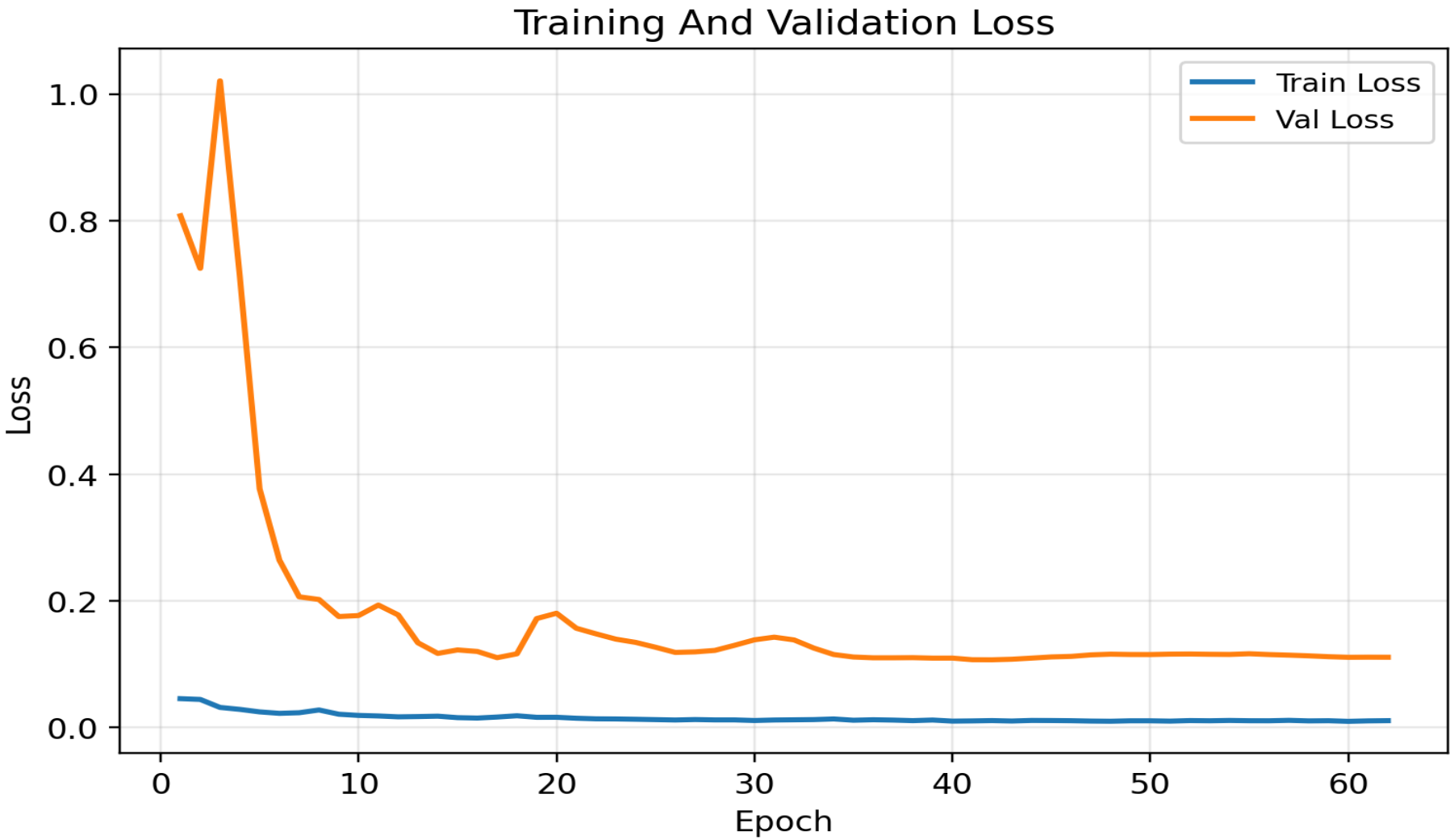
Total train windows before/after balance: 284 / 284

Total val windows before/after balance: 41 / 41

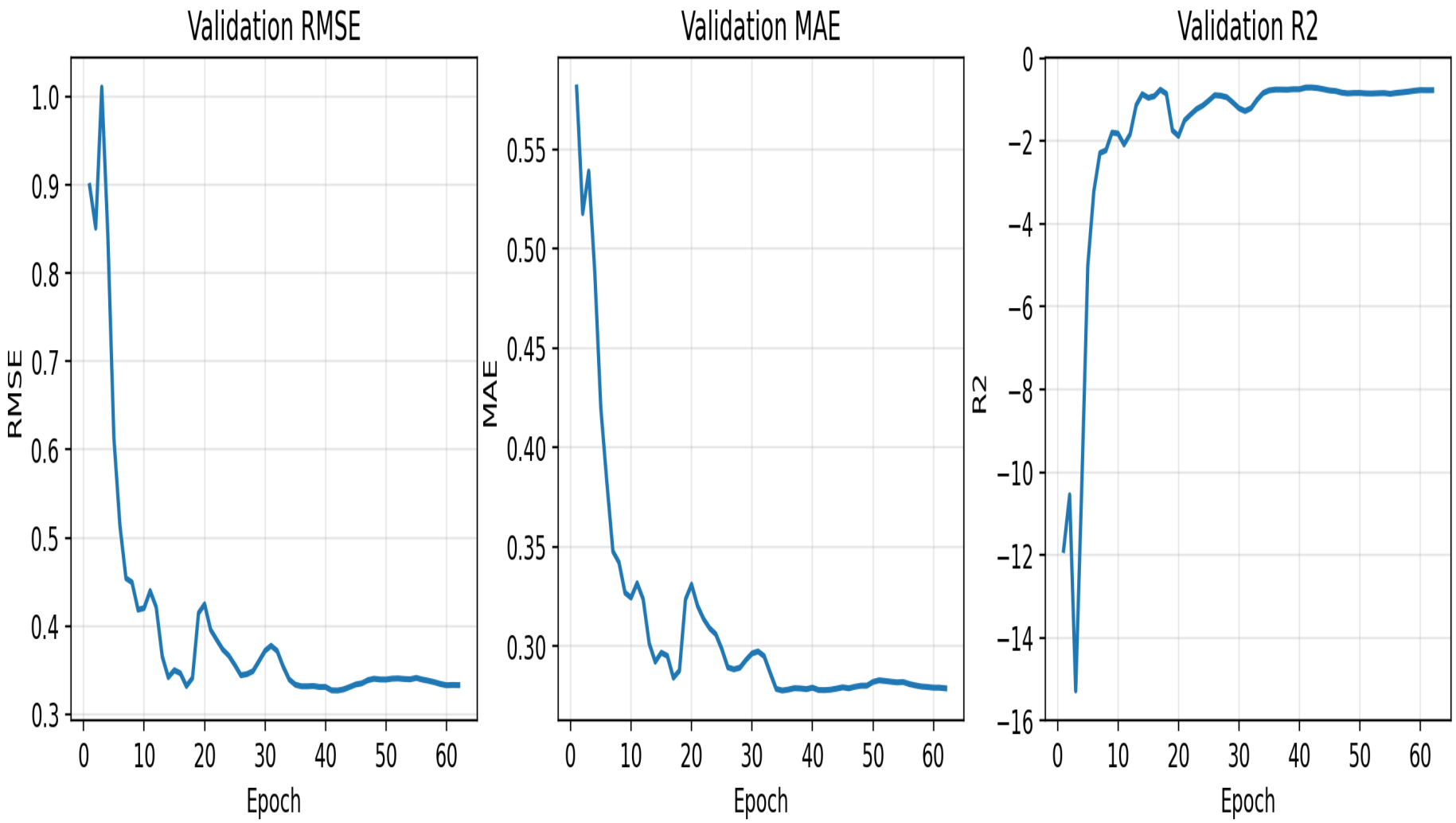
Total test windows: 36

## 2. Core Result Figures

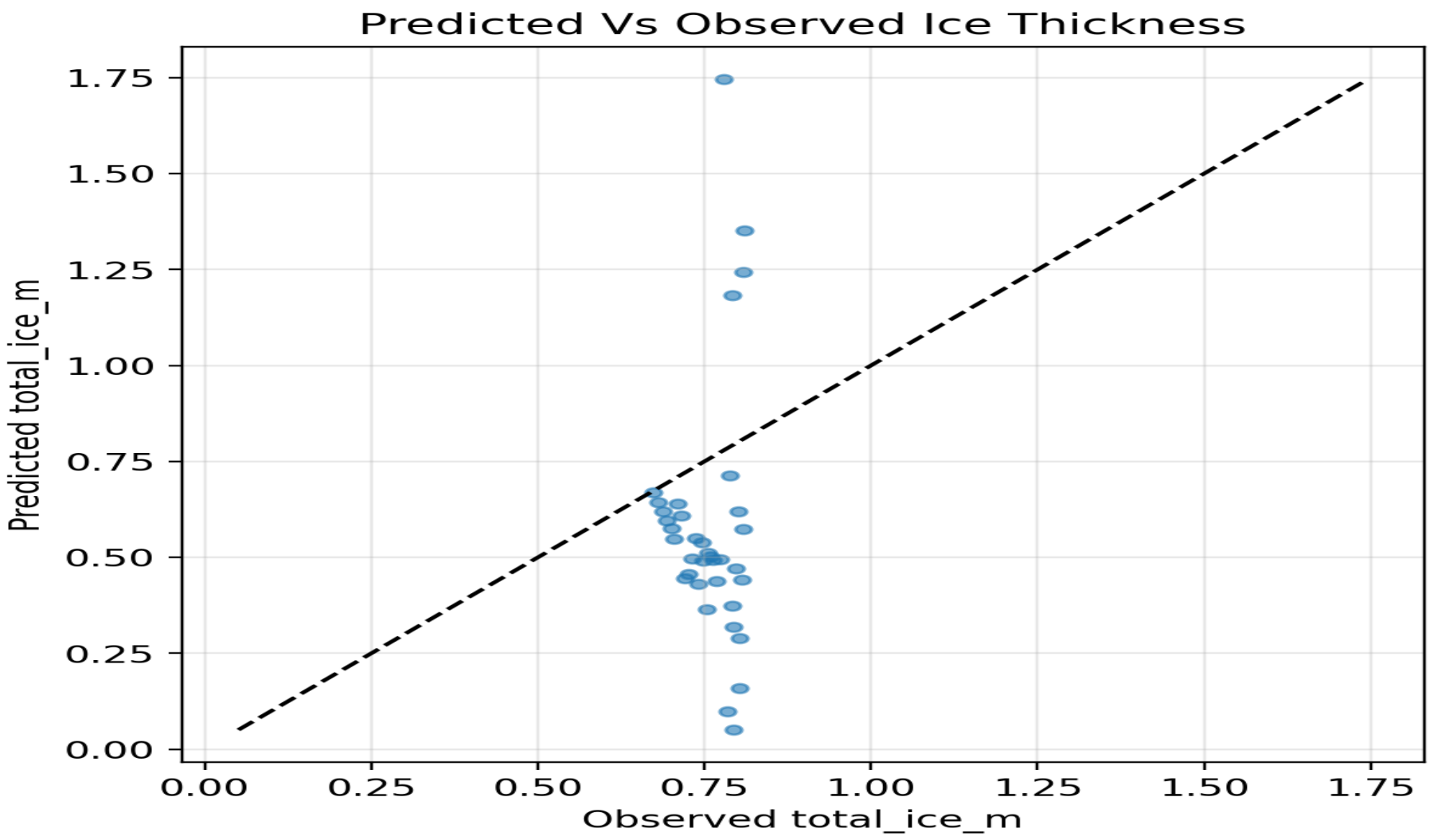
***Training loss curve***



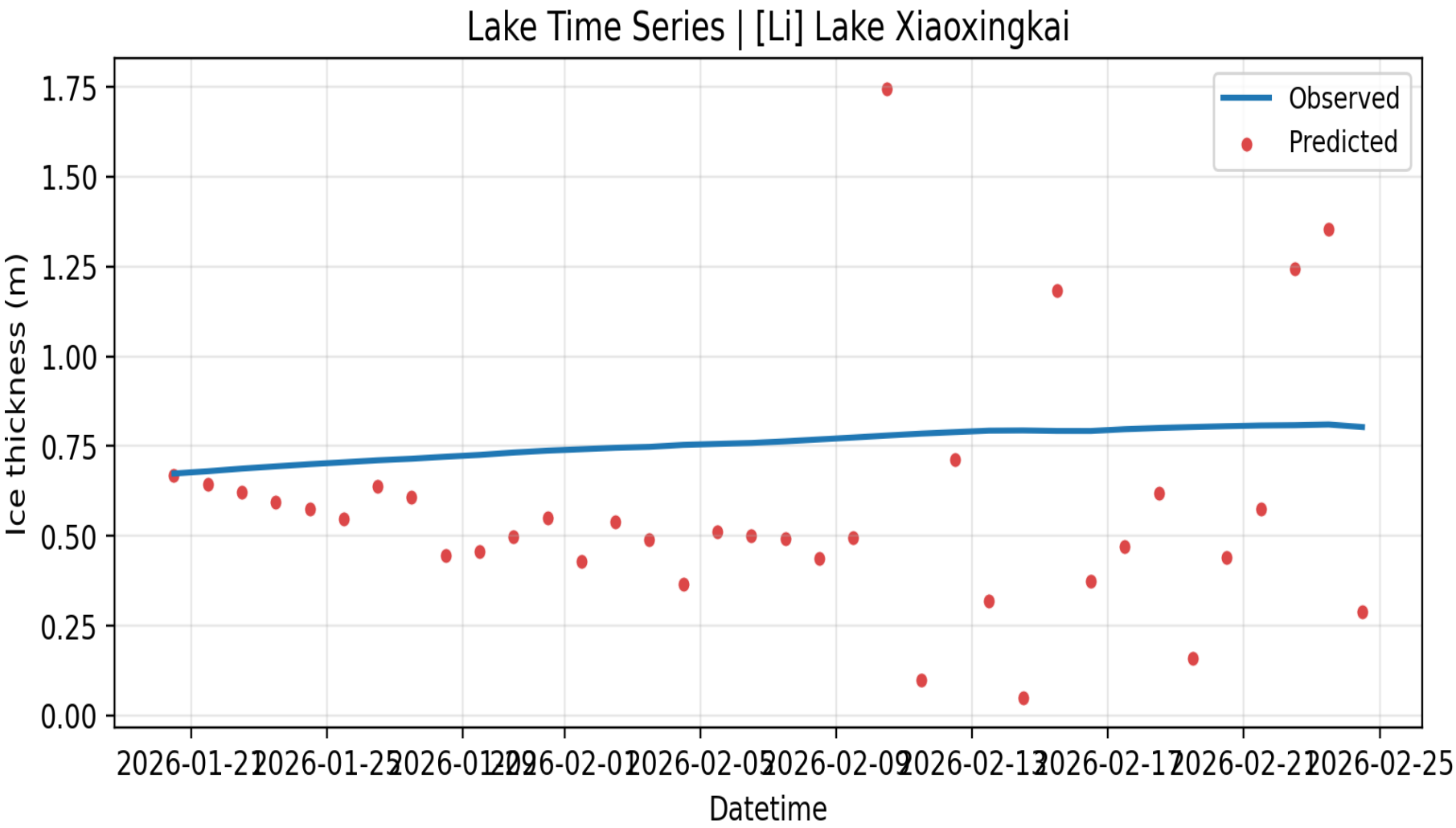

***Validation metric curves***



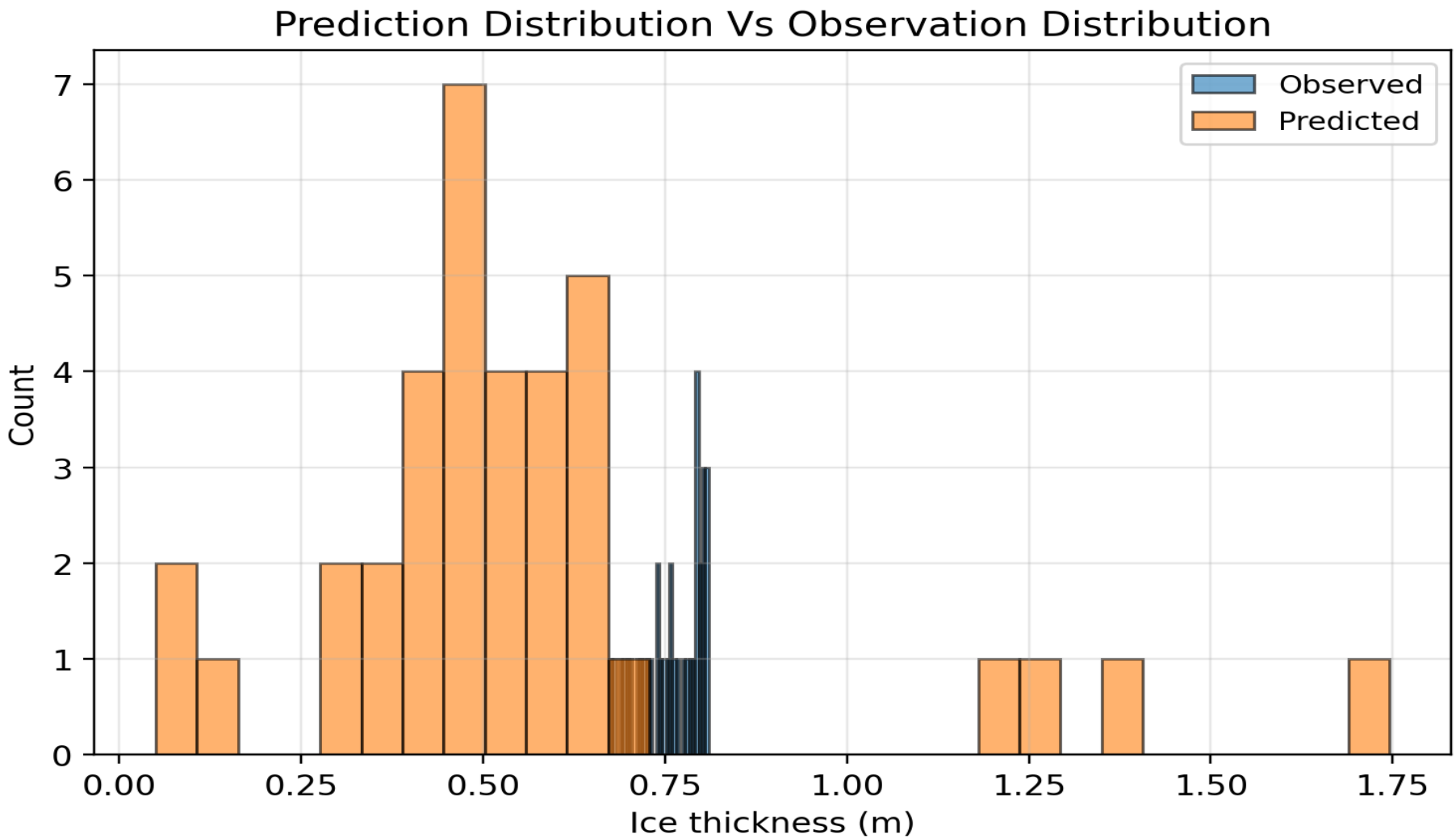

***Evaluation predicted vs observed***




***Xiaoxingkai evaluation time series***




***Prediction vs observation distribution***


Prediction Distribution Vs Observation Distribution
Observed
Predicted
Count
0
1
2
3
4
5
6
7
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
Ice thickness (m)